NEW PHOTOVOLTAIC PITCHED ROOF MOUNTED SYSTEM - 6.450 KW DC/5.235 KW AC

151 BORLAND ST, BROOKSVILLE, FL 34604

NEW PV SYSTEM SPECIFICATIONS

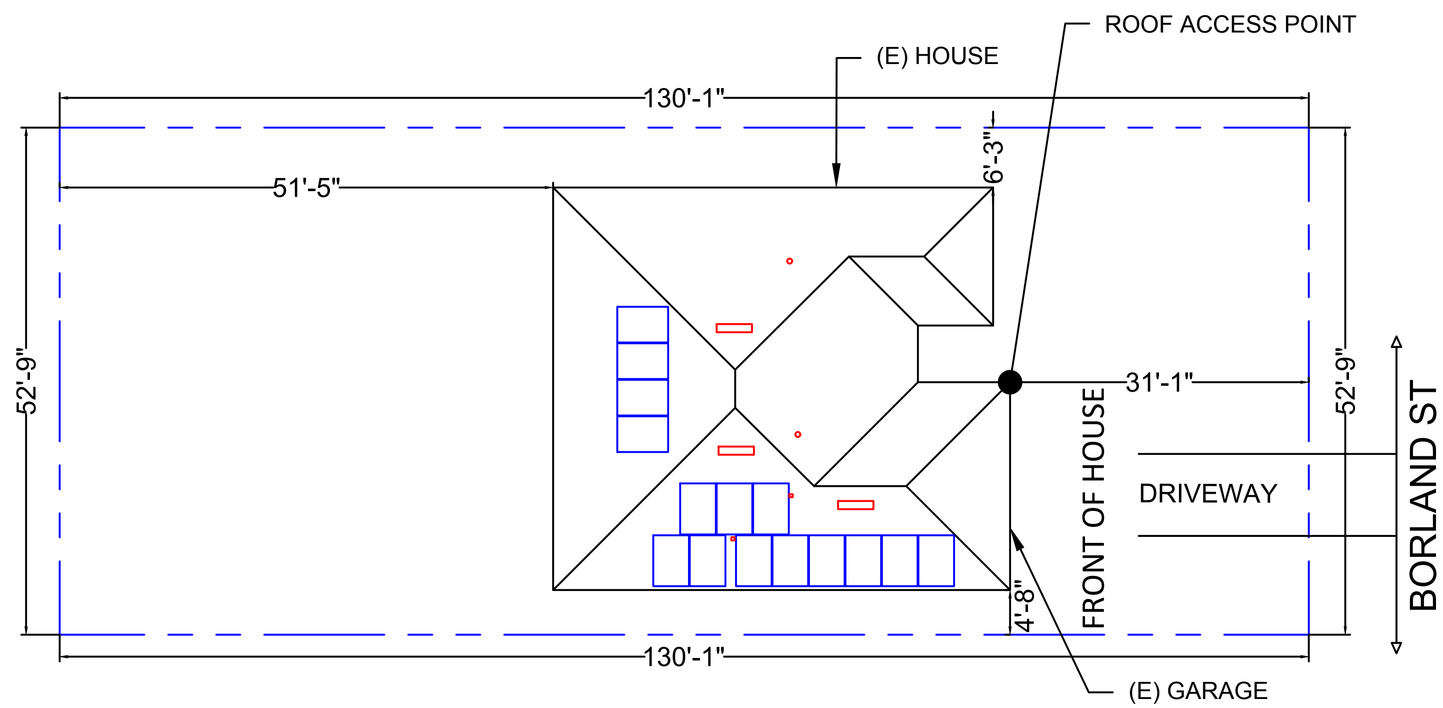
SYSTEM SIZE:	DC SIZE: 6.450 KW DC-(STC)
	AC SIZE: 5.235 KW AC
MODULE:	(15) Q.CELLS Q.TRON BLK M-G2.C1+/AC 430 [430W]
INVERTER:	(15) Q.CELLS Q.MI.349B-G1 [240V] INTEGRATED MICROINVERTERS

APPLICABLE CODES
ALL WORK SHALL CONFORM TO THE FOLLOWING CODES:
2023 FLORIDA BUILDING CODE, 8TH ED
2023 FLORIDA RESIDENTIAL CODE, 8TH ED
2023 FLORIDA FIRE PREVENTION CODE, 8TH ED
2020 NATIONAL ELECTRICAL CODE
AS ADOPTED BY HERNANDO COUNTY

DESIGN CRITERIA
ROOF SURFACE TYPE: ASPHALT SHINGLE
ROOF FRAMING: 2"X4" TRUSS @ 24" OC
BUILDING STORY: ONE STORY
GROUND SNOW LOAD: 3 PSF
WIND SPEED: 140 MPH
WIND EXPOSURE: C
RISK CATEGORY: II

PROJECT NOTES
1.1.1 THIS PHOTOVOLTAIC (PV) SYSTEM SHALL COMPLY WITH THE RELEVANT YEAR OF THE NATIONAL ELECTRICAL CODE (NEC), ALL MANUFACTURER'S LISTING AND INSTALLATION INSTRUCTIONS, AND THE RELEVANT CODES AS SPECIFIED BY THE AUTHORITY HAVING JURISDICTION'S (AHJ) APPLICABLE CODES.
1.1.2 THE UTILITY INTERCONNECTION APPLICATION MUST BE APPROVED AND THE PV SYSTEM MUST BE INSPECTED PRIOR TO OPERATION
1.1.3 ALL PV SYSTEM COMPONENTS; MODULES, UTILITY-INTERACTIVE INVERTERS, AND SOURCE CIRCUIT COMBINER BOXES ARE IDENTIFIED AND LISTED FOR USE IN PHOTOVOLTAIC SYSTEMS AS REQUIRED BY NEC AND OTHER GOVERNING CODES
1.1.4 ALL SIGNAGE TO BE PLACED IN ACCORDANCE WITH LOCAL BUILDING CODE. IF EXPOSED TO SUNLIGHT, IT SHALL BE UV RESISTANT. ALL PLAQUES AND SIGNAGE WILL BE INSTALLED AS REQUIRED BY THE NEC AND AHJ.

SCOPE OF WORK
1.2.1 CONTRACTOR IS RESPONSIBLE FOR THE DESIGN AND SPECIFICATIONS OF THE GRID-TIED PHOTOVOLTAIC SYSTEM. THE CONTRACTOR WILL BE RESPONSIBLE FOR COLLECTION OF EXISTING ONSITE CONDITIONS TO DESIGN, SPECIFY, AND INSTALL THE PITCHED ROOF-MOUNTED PHOTOVOLTAIC SYSTEM DETAILED IN THIS DOCUMENT



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1 PV-01 PROPERTY PLAN
SCALE:1" = 20'-0"

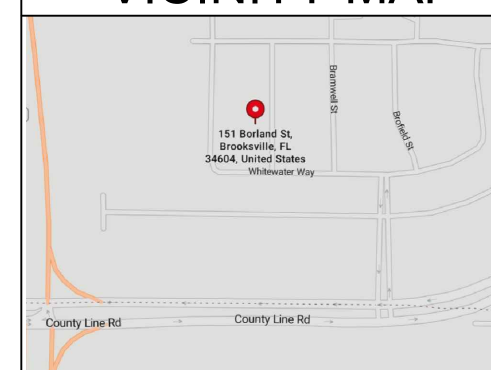
SHEET INDEX

PV-01	COVER PAGE
PV-02	SITE PLAN
PV-03	ATTACHMENT PLAN & DETAILS
PV-04	ELECTRICAL DIAGRAM
PV-05	NOTES
PV-06	WARNING LABELS
PV-07	INSTALLATION RESOURCE
EQUIPMENT DATASHEETS ATTACHED	

LEGEND

- PROPERTY LINE

VICINITY MAP



SATELLITE MAP

CONTRACTOR

UNICITY SOLAR ENERGY LLC

4515 GEORGE RD SUITE #340,
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LIC. NO. - #EC13010036
#CBC1268432

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D. Chad Godwin, PE Chad@godwineng.com

CHAD GODWIN LICENSE No 81360 STATE OF FLORIDA PROFESSIONAL ENGINEER

THIS ITEM HAS BEEN DIGITALLY SIGNED AND SEALED BY DONNIE CHAD GODWIN ON THE DATE ADJACENT TO THE SEAL.

SIGNATURE MUST BE VERIFIED ON ANY ELECTRONIC COPIES.

Digitally signed by Donnie Godwin
Date: 2026.03.25 08:34:35 -05'00'

PROJECT NAME & ADDRESS
DAVESKY ARAGONES
151 BORLAND ST,
BROOKSVILLE, FL 34604

APN #: R35 223 18 3017 0000 0910
AHJ: HERNANDO COUNTY
UTILITY: DUKE ENERGY

SYSTEM DETAILS
6.450 KW DC-(STC) / 5.235 KW AC
(15) Q.CELLS Q.TRON BLK M-G2.C1+/AC 430 [430W]
(15) Q.CELLS Q.MI.349B-G1 [240V] INTEGRATED MICROINVERTERS

REVISIONS

REV	DESCRIPTION	DATE

SHEET TITLE
COVER PAGE

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SHEET NUMBER
PV-01

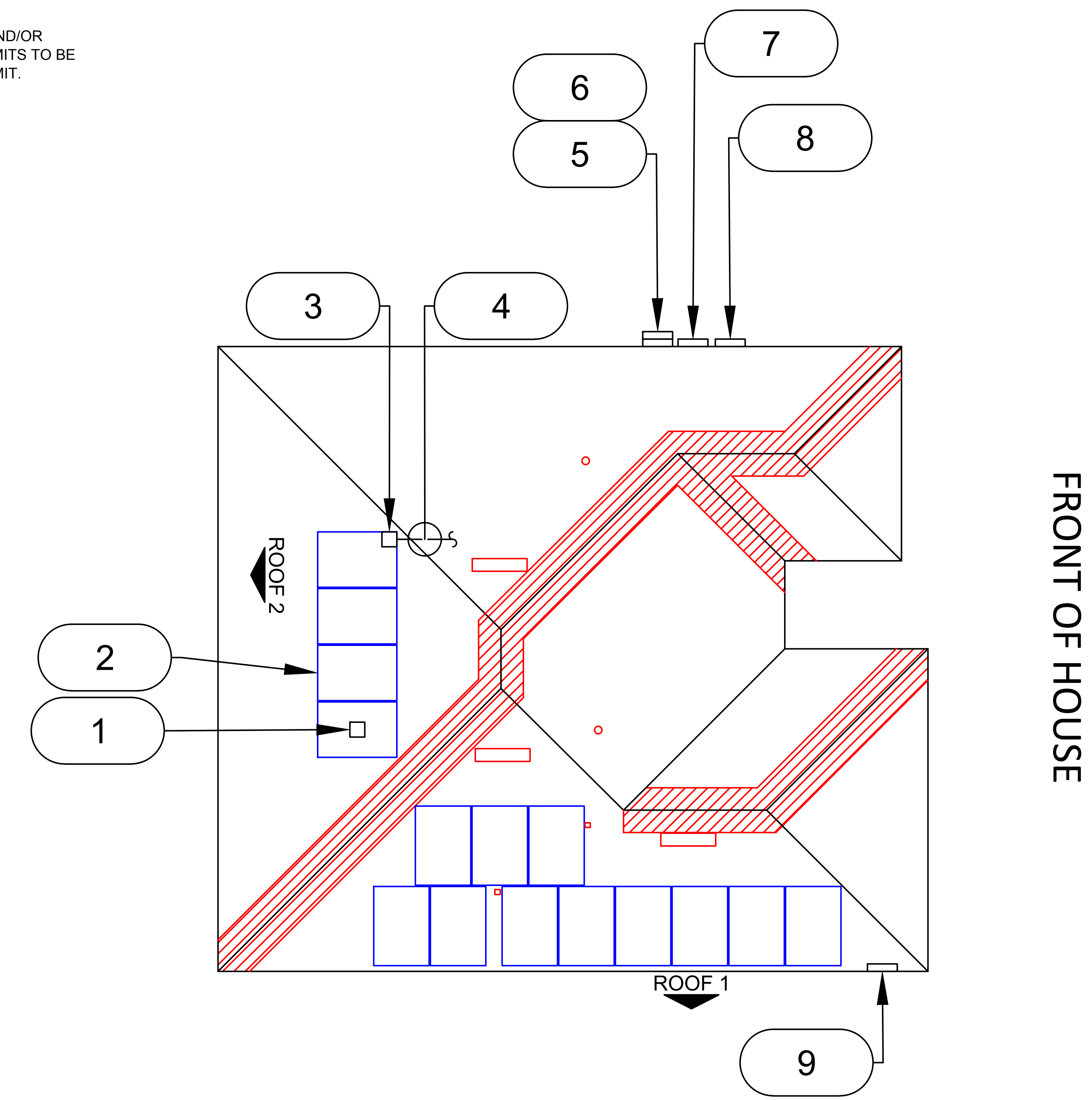
NOTES:

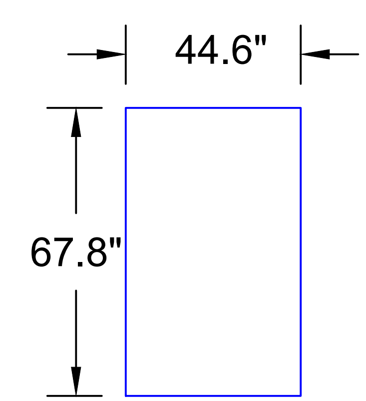
1. ROOF ACCESS POINT SHALL NOT BE LOCATED IN AREAS THAT REQUIRE THE PLACEMENT OF GROUND LADDERS OVER OPENINGS SUCH AS WINDOWS OR DOORS, AND LOCATED AT STRONG POINTS OF BUILDING CONSTRUCTION IN LOCATIONS WHERE THE ACCESS POINT DOES NOT CONFLICT WITH OVERHEAD OBSTRUCTIONS SUCH AS TREE LIMBS, WIRES OR SIGNS.
2. STRUCTURES, PATIO COVERS, AND/OR ADDITIONS BUILT WITHOUT PERMITS TO BE RESOLVED BY A SEPARATE PERMIT.

STRING MOD QTY
STRING #1: 08
STRING #2: 07

PLAN VIEW TOTAL ROOF AREA:	1912.02 FT²
TOTAL PV ARRAY AREA:	314.99 FT²
TOTAL % OF ROOF COVERED BY PV:	16.47%

VISIBLE, LOCKABLE & LABELED AC DISCONNECT LOCATED WITHIN 5FT OF THE UTILITY METER





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PV-02

SITE PLAN

SCALE:1" = 10'-0"

LEGEND

36" 18" FIRE SETBACKS

= MECHANICAL VENT

= FLUE / PLUMBING VENT

1. MICROINVERTER (1 PER MODULE)
2. (15) Q.CELLS Q.TRON BLK M-G2.C1+/AC 430 [430W] MODULES WITH Q.CELLS Q.MI.349B-G1 [240V] INTEGRATED MICROINVERTERS UNDER EACH MODULE
3. JUNCTION BOX (NEMA 3R)
4. CONDUIT RUN; SURFACE MOUNTED (ACTUAL CONDUIT RUNS TO BE DETERMINED IN FIELD)
5. (E) UTILITY METER (UNDERGROUND SERVICE) METER #: 8424839
6. (E) MAIN SERVICE PANEL
7. (N) AC DISCONNECT (NON FUSED)
8. (N) Q.CELLS Q.HOME COMBINER 80 G1
9. (E) SUB SERVICE PANEL

ROOF 1	SLOPE - 20° AZIMUTH - 180° MODULE QTY - 11 TRUSS - 2"X4" @ 24" O.C. SURFACE TYPE- ASPHALT SHINGLE
ROOF 2	SLOPE - 20° AZIMUTH - 270° MODULE QTY - 4 TRUSS - 2"X4" @ 24" O.C. SURFACE TYPE- ASPHALT SHINGLE

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SITE PLAN

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PV-02

DISTRIBUTED LOAD CALCULATIONS

MODULE	Q.CELLS Q.TRON BLK M-G2.C1+/AC 430 [430W]
MODULE WEIGHT	50.59 LBS
MODULE DIMENSIONS (L" x W")	67.8" x 44.6"
TOTAL QTY. OF MODULES	15
TOTAL WEIGHT OF MODULES	758.9 LBS
TYPE OF RACKING	ECOFASTEN CF STD US RAIL AL MILL 171.5" 2012035
TYPE OF ATTACHMENT	ECOFASTEN CF SMART FOOT W/ CLKR AL MILL 2012028
DISTRIBUTED WEIGHT OF RACKING	0.5 PSF
TOTAL WEIGHT OF ARRAY	916.3 LBS
AREA OF MODULE	21.0 SQFT.
TOTAL ARRAY AREA	315.0 SQFT.
DISTRIBUTED LOAD	2.9 PSF
ATTACHMENT QTY	38

NOTE:

1. CONTRACTOR/INSTALLER TO VERIFY COMPATIBILITY OF ANY BRANDS OR PRODUCTS SUBSTITUTED OR USED AS ALTERNATES WITHIN ANY BRAND-SPECIFIC SYSTEMS. CONTRACTOR SHALL SUPPLY AND PRESENT CERTIFICATES OF COMPATIBILITY TO THE BUILDING OFFICIAL UPON INSPECTION AS NEEDED.
2. REFER TO PV MODULE MANUFACTURER'S INSTALLATION INSTRUCTIONS FOR RAIL SPACING SPECIFICATIONS

TRUSS SPAN	ZONE1	ZONE 2	ZONE 3
EXPOSED	48"	48"	24"
NON EXPOSED	48"	48"	48"
MAX CANTILIVER	16"	16"	16"
MAX CANTILIVER= MAX SPAN*(1/3)			

ROOF 1	SLOPE - 20° AZIMUTH - 180° MODULE QTY - 11 TRUSS - 2"X4" @ 24" O.C. SURFACE TYPE- ASPHALT SHINGLE
ROOF 2	SLOPE - 20° AZIMUTH - 270° MODULE QTY - 4 TRUSS - 2"X4" @ 24" O.C. SURFACE TYPE- ASPHALT SHINGLE

LEGEND

- ATTACHMENT POINTS
- RAIL
- STRUCTURAL MEMBER

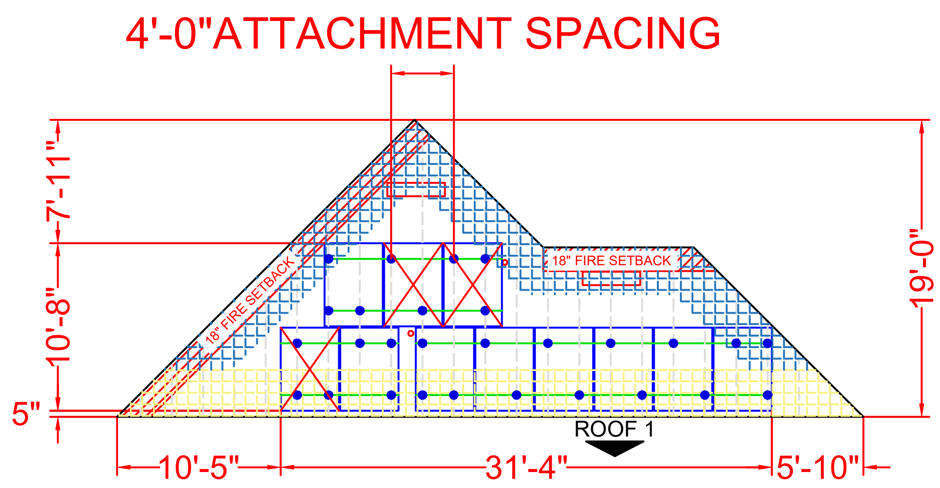
NON-EXPOSED

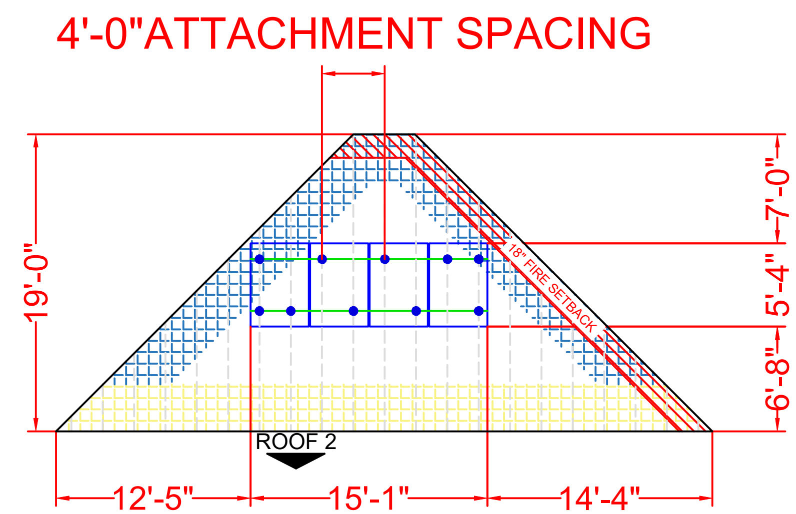
EXPOSED

ZONE 1

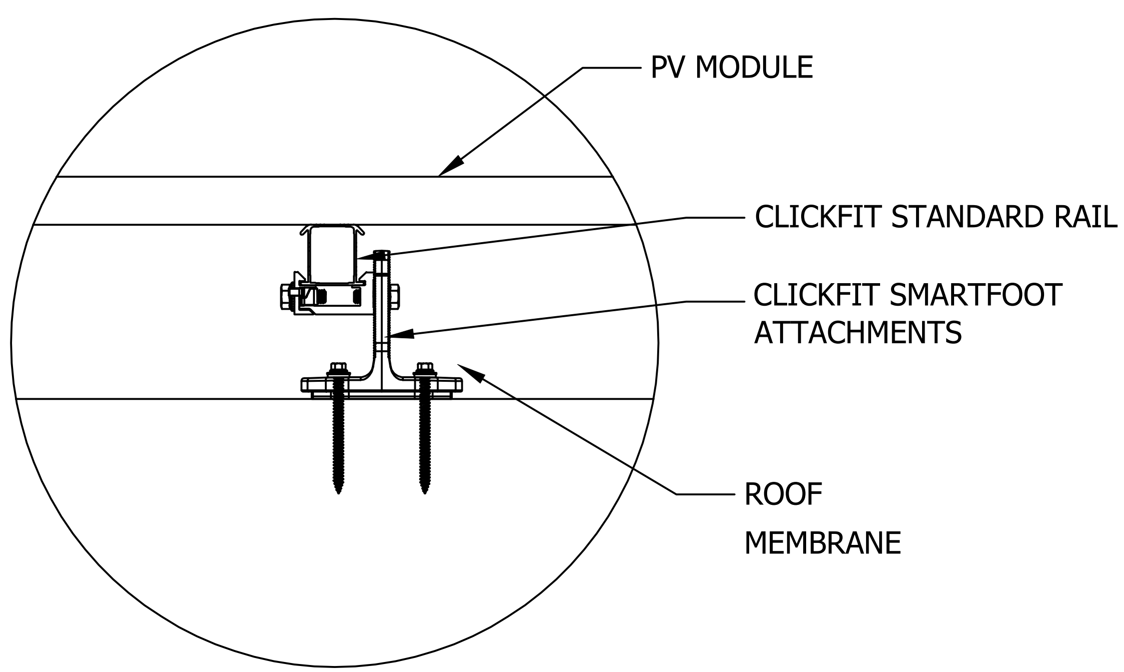
ZONE 2

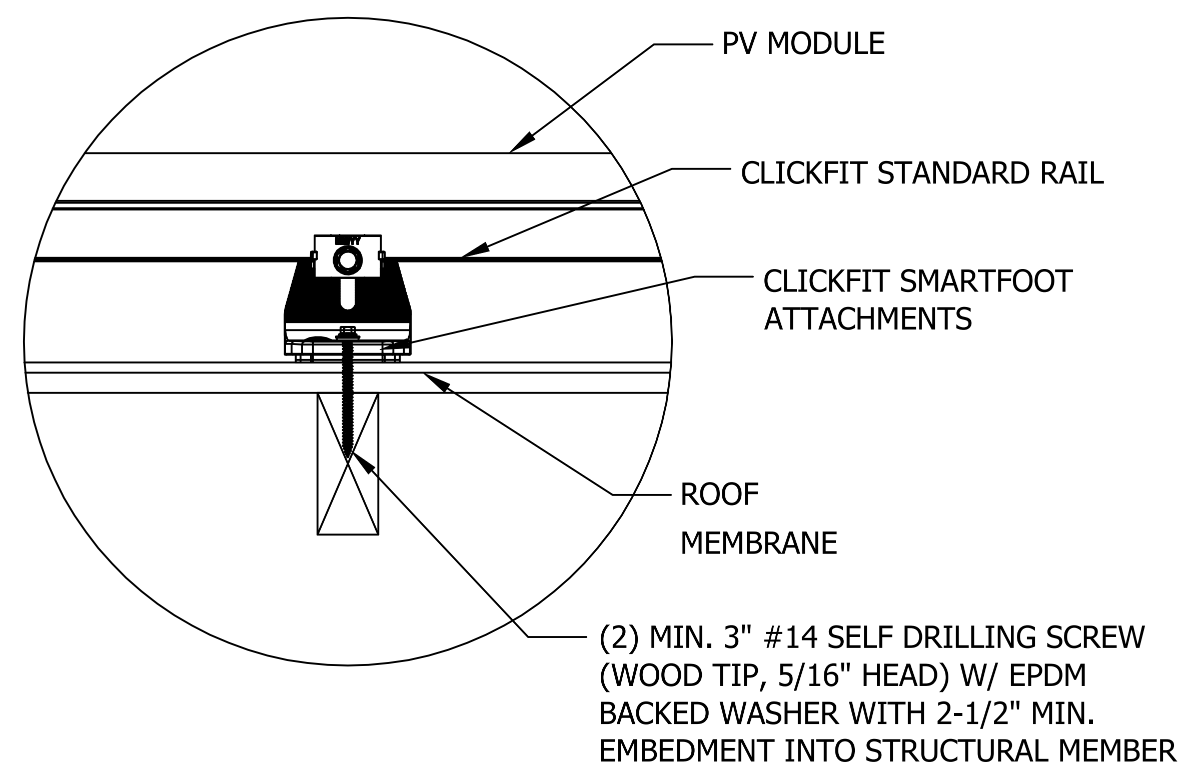
ZONE 3





NOTE: 6" MAXIMUM DISTANCE FROM ROOF SURFACE TO TOP OF PV MODULES





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REVISIONS

REV	DESCRIPTION	DATE

SHEET TITLE

ATTACHMENT PLAN & DETAILS

DRAWN DATE	3/24/2026
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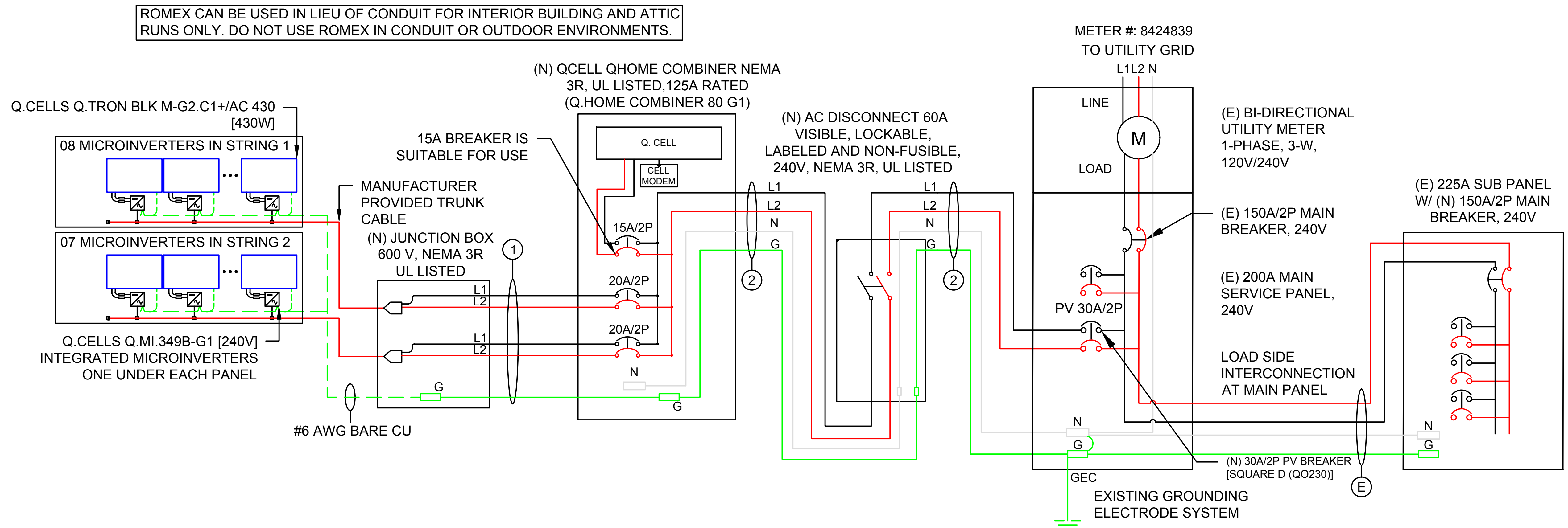
PV-03

INVERTER SPECIFICATIONS	
MANUFACTURER / MODEL #	Q.CELLS Q.MI.349B-G1 [240V] INTEGRATED MICROINVERTERS
POWER RATING	349W
MIN/MAX START VOLTAGE	22V/58V
NOMINAL AC VOLTAGE	240V
MAX CONT. OUTPUT CURRENT	1.45A
MAX MODULES PER STRING	11

SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL #	Q.CELLS Q.TRON BLK M-G2.C1+/AC 430 [430W]
VMP	32.94 V
IMP	13.05 A
VOC	39.32 V
ISC	13.74 A
TEMP. COEFF. VOC	-0.24 %/K

AMBIENT TEMPERATURE SPECIFICATIONS	
RECORD LOW TEMP	-7°C
AMBIENT TEMP (HIGH TEMP 2% AVG.)	34°C
MINIMUM CONDUIT HEIGHT ABOVE ROOF SURFACE	7/8"

VISIBLE, LOCKABLE & LABELED AC DISCONNECT LOCATED WITHIN 5FT OF THE UTILITY METER



DESCRIPTION	FORMULA	RESULT
PV OVERCURRENT PROTECTION NEC 690.9(B)	TOTAL INVERTER OUTPUT CURRENT x 1.25 = 21.75A x 1.25	27.19A (SELECTED PV BREAKER = 30A)
120% RULE FOR BACKFEED BREAKER NEC 705.12	BUS BAR RATING x 1.2 - MCB RATING = MAX ALLOWABLE PV BREAKER 200A x 1.2 - 150A = 90A	SELECTED PV BREAKER <= MAX ALLOWABLE PV BREAKER 30A <= 90A

WIRE ID	EXPECTED WIRE TEMP (°C)	TEMP DERATE (90 °C)	QTY OF CURRENT CARRYING CONDUCTORS	CONDUIT FILL DERATE	MINIMUM CONDUIT SIZE (TBD ON SITE)	WIRE GAUGE & TYPE	CONDUCTOR AMPACITY @ 90°C (A)	CONDUCTOR AMPACITY @ 75°C (A)	REQUIRED CIRCUIT CONDUCTOR AMPACITY (A)	ADJUSTED CONDUCTOR AMPACITY @ 90 °C (A)	NEUTRAL CONDUCTOR SIZE & TYPE	GROUND WIRE SIZE & TYPE
1	34	0.96	4	0.8	3/4"	#10 THWN-2	40	35	14.5	30.72	NONE	#10 THWN-2
2	34	0.96	2	1	3/4"	#10 THWN-2	40	35	27.19	38.4	#10 THWN-2	#10 THWN-2

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ELECTRICAL DIAGRAM

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PV-04

GENERAL NOTES

SITE NOTES
2.1.1 A LADDER WILL BE IN PLACE FOR INSPECTION IN ACCORDANCE WITH OSHA REGULATIONS.
2.1.2 THE PV MODULES ARE CONSIDERED NON-COMBUSTIBLE AND THIS SYSTEM IS A UTILITY INTERACTIVE SYSTEM WITH NO STORAGE BATTERIES.
2.1.3 THE SOLAR PV INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS.
2.1.4 PROPER ACCESS AND WORKING CLEARANCE AROUND EXISTING AND PROPOSED ELECTRICAL EQUIPMENT WILL BE PROVIDED IN ACCORDANCE WITH SECTION NEC 110
2.1.5 ROOF COVERINGS SHALL BE DESIGNED, INSTALLED, AND MAINTAINED IN ACCORDANCE WITH THIS CODE AND THE APPROVED MANUFACTURER'S INSTRUCTIONS SUCH THAT THE ROOF COVERING SERVES TO PROTECT THE BUILDING OR STRUCTURE.

EQUIPMENT LOCATIONS
2.2.1 ALL EQUIPMENT SHALL MEET MINIMUM WORKING CLEARANCES IN ACCORDANCE WITH NEC 110.26.
2.2.2 WIRING SYSTEMS INSTALLED IN DIRECT SUNLIGHT MUST BE RATED FOR EXPECTED OPERATING TEMPERATURE AS SPECIFIED BY 2020 NATIONAL ELECTRICAL CODE.
2.2.3 JUNCTION AND PULL BOXES PERMITTED INSTALLED UNDER PV MODULES IN ACCORDANCE WITH NEC 690
2.2.4 ALL EQUIPMENT SHALL BE INSTALLED ACCESSIBLE TO QUALIFIED PERSONNEL IN ACCORDANCE WITH NEC APPLICABLE CODES.
2.2.5 ALL COMPONENTS ARE LISTED FOR THEIR PURPOSE AND RATED FOR OUTDOOR USAGE WHEN APPROPRIATE.

STRUCTURAL NOTES
2.3.1 RACKING SYSTEM & PV ARRAY WILL BE INSTALLED IN ACCORDANCE WITH THE CODE-COMPLIANT INSTALLATION MANUAL. TOP CLAMPS REQUIRE A DESIGNATED SPACE BETWEEN MODULES, AND RAILS MUST ALSO EXTEND A MINIMUM DISTANCE BEYOND EITHER EDGE OF THE ARRAY/SUBARRAY, IN ACCORDANCE WITH RAIL MANUFACTURER'S INSTALLATION PRACTICES.
2.3.2 JUNCTION BOX WILL BE INSTALLED PER MANUFACTURER'S SPECIFICATIONS. IF ROOF-PENETRATING TYPE, IT SHALL BE FLASHED & SEALED PER LOCAL REQUIREMENTS.
2.3.3 ROOFTOP PENETRATIONS FOR PV RACEWAY WILL BE COMPLETED AND SEALED W/ APPROVED CHEMICAL SEALANT PER CODE BY A LICENSED CONTRACTOR.
2.3.4 ALL PV RELATED ROOF ATTACHMENTS TO BE SPACED NO GREATER THAN THE SPAN DISTANCE SPECIFIED BY THE RACKING MANUFACTURER OR PROFESSIONAL ENGINEERING GUIDANCE.
2.3.5 WHEN POSSIBLE, ALL PV RELATED RACKING ATTACHMENTS WILL BE STAGGERED AMONGST THE ROOF FRAMING MEMBERS.

WIRING & CONDUIT NOTES
2.4.1 ALL CONDUIT AND WIRE WILL BE LISTED AND APPROVED FOR THEIR PURPOSE. CONDUIT AND WIRE SPECIFICATIONS ARE BASED ON MINIMUM CODE REQUIREMENTS AND ARE NOT MEANT TO LIMIT UP-SIZING.
2.4.2 CONDUCTORS SIZED IN ACCORDANCE WITH THE NEC
2.4.3 AC CONDUCTORS TO BE COLORED OR MARKED PER NEC
2.4.4 LISTED OR LABELED EQUIPMENT SHALL BE INSTALLED AND USED IN ACCORDANCE WITH ANY INSTRUCTIONS INCLUDED IN THE LISTING OR LABELING PER NEC

GROUNDING NOTES
2.5.1 GROUNDING SYSTEM COMPONENTS SHALL BE LISTED FOR THEIR PURPOSE, AND GROUNDING DEVICES EXPOSED TO THE ELEMENTS SHALL BE RATED FOR SUCH USE.
2.5.2 PV EQUIPMENT SHALL BE GROUNDED IN ACCORDANCE WITH NEC 690 AND NEC TABLE 250.122.
2.5.3 METAL PARTS OF MODULE FRAMES, MODULE RACKING, AND ENCLOSURES CONSIDERED GROUNDED IN ACCORDANCE WITH NEC 250.
2.5.4 EQUIPMENT GROUNDING CONDUCTORS SHALL BE SIZED IN ACCORDANCE WITH NEC 690 AND INVERTER MANUFACTURER'S INSTALLATION PRACTICES
2.5.5 EACH MODULE WILL BE GROUNDED AS SHOWN IN MANUFACTURER DOCUMENTATION AND APPROVED BY THE AHJ.
2.5.6 THE GROUNDING CONNECTION TO A MODULE SHALL BE ARRANGED SUCH THAT THE REMOVAL OF A MODULE DOES NOT INTERRUPT A GROUNDING CONDUCTOR TO ANOTHER MODULE.
2.5.7 GROUNDING AND BONDING CONDUCTORS, IF INSULATED, SHALL BE COLORED GREEN OR MARKED GREEN IF #4 AWG OR LARGER PER NEC 250
2.5.8 THE GROUNDING ELECTRODE SYSTEM COMPLIES WITH NEC 690 AND NEC 250. IF EXISTING SYSTEM IS INACCESSIBLE, OR INADEQUATE, A GROUNDING ELECTRODE SYSTEM PROVIDED IN ACCORDANCE WITH NEC 250, NEC 690 AND THE AHJ.
2.5.9 GROUND-FAULT DETECTION SHALL COMPLY WITH NEC 690 TO REDUCE FIRE HAZARDS

DISCONNECTION AND OVERCURRENT PROTECTION NOTES
2.6.1 DISCONNECTING SWITCHES SHALL BE WIRED SUCH THAT WHEN THE SWITCH IS OPENED THE CONDUCTORS REMAINING ENERGIZED ARE CONNECTED TO THE TERMINALS MARKED “LINE SIDE” (TYPICALLY THE UPPER TERMINALS).
2.6.2 DISCONNECTS TO BE ACCESSIBLE TO QUALIFIED UTILITY PERSONNEL, BE LOCKABLE, AND BE A VISIBLE-BREAK SWITCH
2.6.3 PV SYSTEM CIRCUITS INSTALLED ON OR IN HABITABLE BUILDINGS SHALL INCLUDE A RAPID SHUTDOWN FUNCTION TO REDUCE SHOCK HAZARD FOR EMERGENCY RESPONDERS IN ACCORDANCE WITH 690
2.6.4 ALL OCPD RATINGS AND TYPES SPECIFIED ACCORDING TO NEC 690 AND 240.
2.6.5 INVERTER ON-GRID BRANCHES SHALL BE CONNECTED TO A SINGLE BREAKER OR GROUPED FUSE DISCONNECT(S) IN ACCORDANCE WITH NEC 110.
2.6.6 IF REQUIRED BY THE AHJ, SYSTEM WILL INCLUDE ARC-FAULT CIRCUIT PROTECTION IN ACCORDANCE WITH NEC 690.11 AND UL1699B.

INTERCONNECTION NOTES
2.7.1 LOAD SIDE INTERCONNECTION SHALL BE IN ACCORDANCE WITH NEC 705.
2.7.2 THE SUM OF THE UTILITY OCPD AND INVERTER CONTINUOUS OUTPUT MAY NOT EXCEED 120 PERCENT OF BUSBAR RATING PER NEC 705.
2.7.3 THE SUM OF 125 PERCENT OF THE POWER SOURCE(S) OUTPUT CIRCUIT CURRENT AND THE RATING OF THE OVERCURRENT DEVICE PROTECTING THE BUSBAR SHALL NOT EXCEED 120 PERCENT OF THE AMPACITY OF THE BUSBAR, PV DEDICATED BACKFEED BREAKERS MUST BE LOCATED OPPOSITE END OF THE BUS FROM THE UTILITY SOURCE OCPD IN ACCORDANCE WITH NEC 705.
2.7.4 AT MULTIPLE ELECTRIC POWER SOURCES OUTPUT COMBINER PANEL, TOTAL RATING OF ALL OVERCURRENT PROTECTION DEVICES SHALL NOT EXCEED AMPACITY OF BUSBAR. HOWEVER, THE MAIN OVERCURRENT PROTECTION DEVICE MAY BE EXCLUDED IN ACCORDANCE WITH NEC 705.
2.7.5 FEEDER TAP INTERCONNECTION (LOAD SIDE) IN ACCORDANCE WITH NEC 705.
2.7.6 SUPPLY SIDE TAP INTERCONNECTION IN ACCORDANCE WITH TO NEC 705 WITH SERVICE ENTRANCE CONDUCTORS IN ACCORDANCE WITH NEC 230.
2.7.7 BACKFEEDING BREAKER FOR ELECTRIC POWER SOURCES OUTPUT IS EXEMPT FROM ADDITIONAL FASTENING PER NEC 705.

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NOTES

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PV-05

WARNING

ELECTRICAL SHOCK HAZARD

TERMINALS ON THE LINE AND LOAD SIDES MAY BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION: COMBINER PANEL, AC DISCONNECT, POINT OF INTERCONNECTION
PER CODE: NEC 706.15(C)(4), NEC 690.13(B)

WARNING

TURN OFF PHOTOVOLTAIC AC DISCONNECT PRIOR TO WORKING INSIDE PANEL

LABEL LOCATION: COMBINER PANEL(S), MAIN SERVICE DISCONNECT
PER CODE: NEC 110.27(C), OSHA 1910.145(f)(7)

PHOTOVOLTAIC SYSTEM AC DISCONNECT
RATED AC OUTPUT CURRENT: 21.75 A
NOMINAL OPERATING AC VOLTAGE: 240 V

LABEL LOCATION: AC DISCONNECT/POINT OF INTERCONNECTION
PER CODE: NEC 690.54

WARNING DUAL POWER SOURCE
SECOND SOURCE IS PHOTOVOLTAIC SYSTEM

LABEL LOCATION: MAIN SERVICE DISCONNECT, PRODUCTION/NET METER
PER CODE: NEC 690.59, 705.12(C)

PV SYSTEM

DISCONNECT

LABEL LOCATION: AC DISCONNECT
PER CODE: NEC 690.13(B)

WARNING

THIS EQUIPMENT FED BY MULTIPLE SOURCES:
TOTAL RATING OF ALL OVERCURRENT DEVICES EXCLUDING MAIN POWER SUPPLY SHALL NOT EXCEED AMPACITY OF BUSBAR

LABEL LOCATION: AC DISCONNECT
PER CODE: NEC 705.12(B)(3)(3)

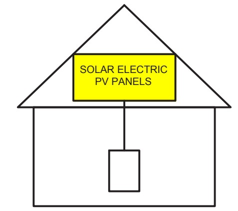
WARNING

POWER SOURCE OUTPUT CONNECTION. DO NOT RELOCATE THIS OVERCURRENT DEVICE.

LABEL LOCATION: POINT OF INTERCONNECTION
PER CODE: NEC 705.12(B)(3)(2)

SOLAR PV SYSTEM EQUIPPED WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN SWITCH TO THE "OFF" POSITION TO SHUT DOWN PV SYSTEM AND REDUCE SHOCK HAZARD IN THE ARRAY



LABEL LOCATION: MAIN SERVICE DISCONNECT
PER CODE: NEC 690.56(C)

MAIN PHOTOVOLTAIC SYSTEM DISCONNECT

LABEL LOCATION: MAIN SERVICE DISCONNECT, UTILITY METER
PER CODE: NEC 690.13(B)

RAPID SHUTDOWN SWITCH FOR SOLAR PV SYSTEM

LABEL LOCATION: RSD INITIATION DEVICE, AC DISCONNECT
PER CODE: NEC 690.56(C)(2)

DO NOT DISCONNECT UNDER LOAD

LABEL LOCATION: MAIN SERVICE DISCONNECT
PER CODE: NEC 690.15(B) & NEC 690.33(D)(2)

EMERGENCY CONTACT
UNICITY SOLAR ENERGY LLC
855-765-7273
FOR IMMEDIATE ASSISTANCE CALL 911

LABEL LOCATION: POINT OF INTERCONNECTION
PER CODE: NEC 705.10

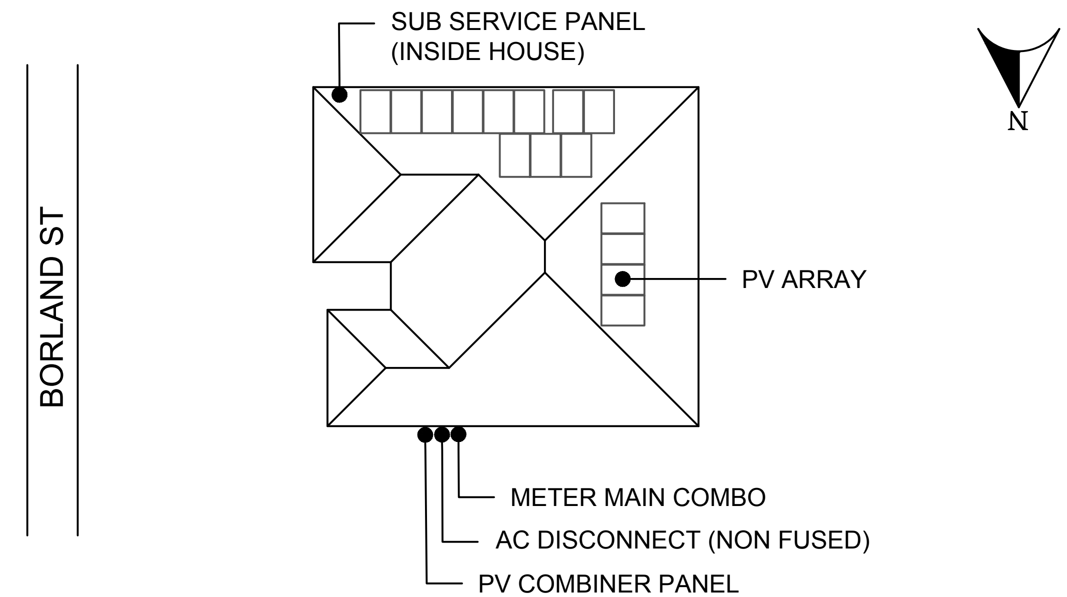
CAUTION
PHOTOVOLTAIC SYSTEM CIRCUIT IS BACKFED

LABEL LOCATION: MAIN SERVICE DISCONNECT
PER CODE: NEC 705.12(D), NEC 690.59

CAUTION

MULTIPLE SOURCES OF POWER.
POWER TO THIS BUILDING IS ALSO SUPPLIED FROM THE FOLLOWING SOURCES WITH DISCONNECTS LOCATED AS SHOWN:

ADDRESS: 151 BORLAND ST, BROOKSVILLE, FL 34604



CONTRACTOR

UNICITY SOLAR ENERGY LLC

4515 GEORGE RD SUITE #340,
TAMPA, FL 33634
PHONE - 855-765-7273
LIC. NO. - #EC13010036
#CBC1268432

Godwin Engineering & Design, LLC
8378 Foxtail Loop Pensacola, FL 32526
D. Chad Godwin, PE Chad@godwineng.com

D. CHAD GODWIN LICENSE No 81360 STATE OF FLORIDA PROFESSIONAL ENGINEER

PROJECT NAME & ADDRESS
DAVESKY ARAGONES
151 BORLAND ST,
BROOKSVILLE, FL 34604

APN #: R35 223 18 3017 0000 0910
AHJ: HERNANDO COUNTY
UTILITY: DUKE ENERGY

SYSTEM DETAILS
6.450 KW DC-(STC) / 5.235 KW AC
(15) Q.CELLS Q.TRON BLK M-G2.C1+/AC 430 [430W]
(15) Q.CELLS Q.MI.349B-G1 [240V] INTEGRATED MICROINVERTERS

REVISIONS

REV	DESCRIPTION	DATE

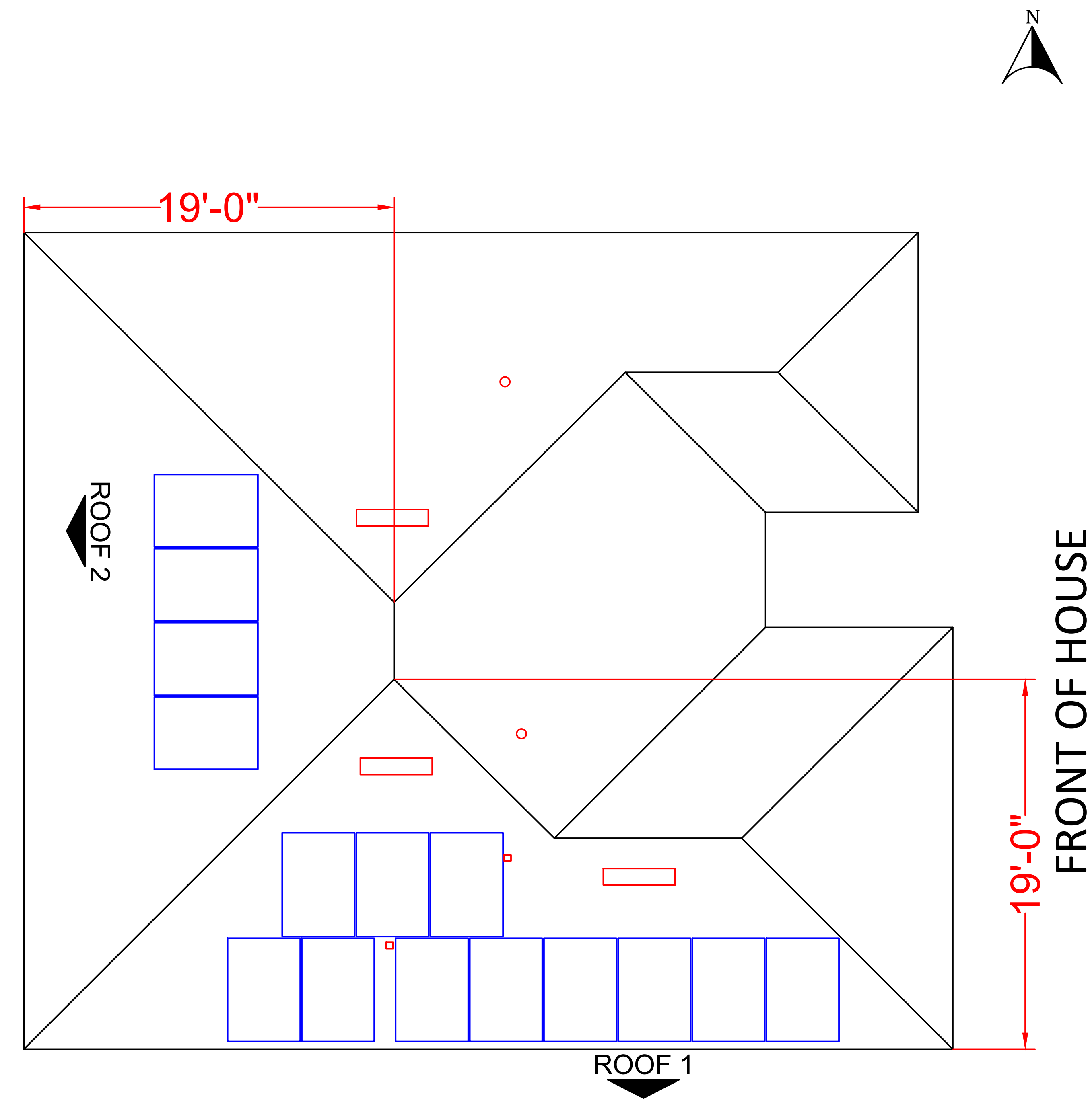
SHEET TITLE
WARNING LABELS

DRAWN DATE	3/24/2026
DRAWN BY	OSS

SHEET NUMBER
PV-06

	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

REFERENCE ONLY



**DIMENSIONS ARE 2D (FLAT)

CONTRACTOR

Unicity Solar Energy

UNICITY SOLAR ENERGY LLC

4515 GEORGE RD SUITE #340,
TAMPA, FL 33634
PHONE - 855-765-7273
LIC. NO. - #EC13010036
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DAVESKY ARAGONES
151 BORLAND ST,
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APN #: R35 223 18 3017 0000 0910
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SYSTEM DETAILS
6.450 KW DC-(STC) / 5.235 KW AC
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(15) Q.CELLS Q.MI.349B-G1 [240V] INTEGRATED MICROINVERTERS

REVISIONS

REV	DESCRIPTION	DATE

SHEET TITLE
INSTALLATION RESOURCE

DRAWN DATE	3/24/2026
DRAWN BY	OSS

SHEET NUMBER
PV-07

Solar Simplified.

Q.TRON AC

Q.TRON BLK M-G2+/AC
Q.TRON BLK M-G2.C1+/AC
Q.TRON BLK M-G2.H1+/AC
Q.TRON BLK M-G2.XY+/AC

Q.TRON AC

AC module powered by Q.ANTUM NEO Technology

Monitoring and Control

- The Q.OMMAND PRO App enables installers to monitor system performance at the module level, while the user-friendly Q.OMMAND HOME App provides homeowners with real-time PV production insights.

Streamlined Installation and Product Management

- Fast installation enabled by integrated Qcells microinverter
- Improved inventory management enabled by reduced SKU counts and one complete module and MLPE solution
- Seamlessly couples with Qcells' residential energy storage system to form one complete Q.HOME SMART system

Superior Module Performance

- Q.TRON AC is powered by Q.ANTUM NEO Technology, boosting module efficiency up to 22.5 % which results in more power production over time.

Top Quality Customer Support

- While the detachable microinverter simplifies on-site maintenance, Qcells' first-class customer support offers rapid system troubleshooting.

Dependably Backed by One Warrantor

- 25-year product and performance warranty with an integrated module and microinverter solution from Qcells.

Includes Domestic Content

- Q.TRON BLK M-G2.XY+/AC contains U.S. manufactured components which can contribute to qualifying for the 10 % domestic content bonus for applicable investment and production tax credits.[1]
- Module and microinverter both assembled in the USA by America's No.1 residential module manufacturer

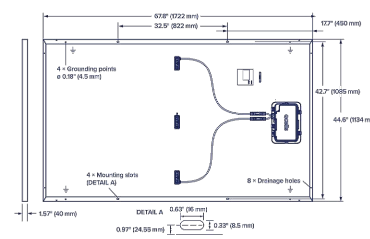
[1] This statement should not be relied on as tax advice and is subject to change based on changes made to the applicable rules and regulations. Please consult a qualified tax professional for specific guidance.

Description

The Q.TRON AC SERIES is a N-type TOPCon PV module with an integrated microinverter. The module, with its embedded microinverter, provides optimized power output while also acting as a rapid shutdown compliant solution for optimal system safety. The solution includes a microinverter, DC cables and a junction box, enabling a streamlined installation experience.

Mechanical Specification

Format	67.8 in × 44.6 in × 1.57 in (including frame) (1722 mm × 1134 mm × 40 mm)
Weight	50.59 lbs (22.95 kg)
Front Cover	0.13 in (3.2 mm) thermally pre-stressed ARC solar glass
Back Cover	Composite film
Frame	Black anodized aluminum
Cell	6 × 18 monocrystalline Q.ANTUM NEO solar half cells
Junction Box	2.09-3.98 in × 1.26-2.36 in× 0.59-0.71 in (53-101 mm × 32-60 mm × 15-18 mm), Protection class IP68, with bypass diodes
Microinverter	9.61 in × 5.79 in × 1.17 in (244 mm × 147 mm × 29.6 mm), Protection class IP67 / NEMA Type 6
DC Cable	4 mm² Solar cable; (+) ≥25.8 in (655 mm), (−) ≥25.2 in (640 mm)
DC Connector	Stäubli MC4; IP68



AC Output Electrical Characteristics

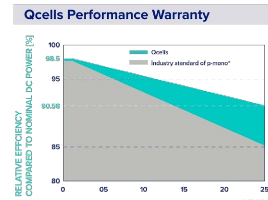
Q.MI.349B-G1 (Model Name)					
Peak Output Power	[VA]	366	Power Factor (adjustable)		0.85 leading...0.85 lagging
Max Continuous Output Power	[VA]	349	Max. number of AC Modules per Q.HOME COMBINER 80 G1	[ea]	44 (Q.HOME COMBINER CB : Max 4)
Nominal (L-L) Voltage / Range	[V]	240 / 211 to 264	Max Units per 20 A (L-L) Branch Circuit	[ea]	11
Nominal Rated Output Current	[A]	1.45	Total Harmonic Distortion	[%]	<5
Nominal Frequency / Range	[Hz]	60 / 59.3 to 60.5	Overvoltage Class AC Port		III
Extended Frequency Range	[Hz]	50 to 66	Night-Time Power Consumption	[mW]	60
Power Factor at Rated Power		1.0	CEC Efficiency	[%]	97

DC Power Electrical Characteristics

Power Class				420	425	430	435	440
MINIMUM PERFORMANCE AT STANDARD TEST CONDITIONS, STC[1] (POWER TOLERANCE +5 W / −0 W)								
Minimum	Power at MPP[1]	P_{MPP}	[W]	420	425	430	435	440
	Short Circuit Current[1]	I_{SC}	[A]	13.58	13.66	13.74	13.82	13.90
	Open Circuit Voltage[1]	V_{OC}	[V]	38.75	39.03	39.32	39.60	39.88
	Current at MPP	I_{MPP}	[A]	12.91	12.98	13.05	13.13	13.20
	Voltage at MPP	V_{MPP}	[V]	32.54	32.74	32.94	33.14	33.33
	Efficiency[1]	η	[%]	≥21.5	≥21.8	≥22.0	≥22.3	≥22.5
MINIMUM PERFORMANCE AT NORMAL OPERATING CONDITIONS, NMOT[2]								
Minimum	Power at MPP	P_{MPP}	[W]	318.2	322.0	325.8	329.5	333.3
	Short Circuit Current	I_{SC}	[A]	10.94	11.00	11.07	11.14	11.20
	Open Circuit Voltage	V_{OC}	[V]	36.77	37.04	37.31	37.58	37.84
	Current at MPP	I_{MPP}	[A]	10.15	10.21	10.27	10.33	10.38
	Voltage at MPP	V_{MPP}	[V]	31.33	31.53	31.72	31.92	32.11

[1] Measurement tolerances P_{MPP} ±3%; I_{SC}, V_{OC} ±5% at STC: 1000 W/m², 25±2 °C, AM 1.5 according to IEC 60904-3 • [2] 800 W/m², NMOT, spectrum AM 1.5

Qcells Performance Warranty

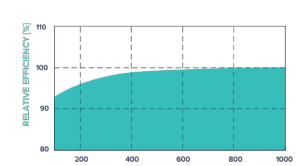


At least 98.5 % of nominal DC power during first year. Thereafter max. 0.33 % degradation per year. At least 95.53 % of nominal DC power up to 10 years. At least 90.58 % of nominal DC power up to 25 years.

All data within measurement tolerances. Full warranties in accordance with the warranty terms of the Qcells sales organization of your respective country.

*Standard terms of guarantee for the 5 PV companies with the highest production capacity in 2021 (February 2021)

Performance at low Irradiance



Typical module performance under low irradiance conditions in comparison to STC conditions (25 °C, 1000 W/m²).

Temperature Coefficients

Temperature Coefficient of I_{SC}	α	[% / K]	+0.04	Temperature Coefficient of V_{OC}	β	[% / K]	−0.24
Temperature Coefficient of P_{MPP}	γ	[% / K]	−0.29	Nominal Module Operating Temperature	NMOT	[°F]	109 ± 5.4 (43 ± 3 °C)

Properties for System Design

Maximum System Voltage	V_{SYS}	[V]	1000	PV Module Classification	Class II
Maximum Series Fuse Rating		[A DC]	25	Fire Rating Based on ANSI / UL 61730	C / TYPE 2
Max. Design Load, Push / Pull[3]		[lbs / ft²]	113 (5400 Pa) / 75 (3600 Pa)	Permitted Module Temperature on Continuous Duty[2]	−40 °F up to +140 °F (−40 °C up to +60 °C)
Max. Test Load, Push / Pull[3]		[lbs / ft²]	169 (8100 Pa) / 113 (5400 Pa)	Storage Temperature Range[2]	−4 °F up to +113 °F (−20 °C up to +45 °C)

[2] According to the Q.MI.349B-G1, the maximum temperature is stated as "60 °C (+140 °F)", but the maximum temperature of the connected DC module is up to "+158 °F (+70 °C)".
[3] See Installation Manual

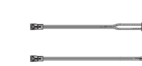
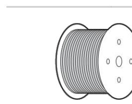
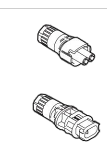
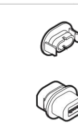
Qualifications and Certificates

Base DC module (Q.TRON BLK M-G2.Y+ solar module series, where "Y" can be any letter between A to W.)
UL 61730-1 & UL 61730-2, CE-compliant;
IEC 61215:2021; IEC 61730:2023.

Qcells Microinverter (Q.MI.349B-G1)
This product is UL listed as PV Rapid Shut Down Equipment
UL1741, UL 1741SA, UL 1741SB, CSA C22.2 No 107.

AC Module (Q.TRON BLK M-G2.XY+/AC solar module series, where "X" can be any letter between A to W and "Y" can be any number between 1 to 9.)
UL 1741, CSA C22.2 No. 107, IEEE E1547.

Accessories (Additional parts, not included in AC module package)

Model		Category
 UL9703 E493181	CAS-HQ-SH-650 CAS-HQ-SH-800 CAS-HQ-SH-900 CAS-HQ-LO-1000 CAS-HQ-LO-1300 CAS-HQ-LO-1400	Short AC cable (L = 650 / 800 / 900 mm): short cable. Intended for connecting portrait oriented PV modules. Long AC cable (L = 1000 / 1300 / 1400 mm): long cable. Intended for connecting landscape oriented PV modules.
 UL3003 E533140	CAB-HQ-KIT-200	AC Cable (Raw) : 200 m cable without AC connector for the free design of AC PV installation. - Detail components : 200 meter (656 ft)
 UL6703 E479328	CON-HQ-KIT-20	AC Connector : To assemble the AC cable (CAB-HQ-KIT-200) by installer themselves. - Detail components : 20pcs Female + 20pcs Male
 UL9703 E493181	ECAP-HQ-KIT-20	End Cap : To close the end of AC cable. - Detail components : 20pcs Female + 20pcs Male
 UL9703 E493181	UNT-HQ-TOOL-G1	AC cable and DC cable Unlocking Tool

Find product recycling details at QR code above

Qcells pursues minimizing paper output in consideration of the global environment.

Note: Installation instructions must be followed. Contact our technical service for further information on approved installation of this product.

Hanwha Q CELLS America Inc. 300 Spectrum Center Drive, Suite 500, Irvine CA, 92618 USA | TEL 1-(888) 249-7750 | EMAIL na.support@qcells.com | WEB www.qcells.com/us

Solar Simplified.

Q.HOME COMBINER

Q.HOME COMBINER 80 G1

Q.HOME COMBINER

Combiner Box of the Q.HOME SMART Residential Energy Solution

Flexible

Network connectivity with Wi-Fi, Ethernet or Cellular

Consolidated

80 A total PV and storage branch circuits

Robust

NRTL-certified NEMA type 3R enclosure rated for outdoor installation

Streamlined

Pre-installed revenue grade production meter, with a pair of slim clamp CTs for consumption metering

Reliable

5 Warranty

- 5-year warranty
- Automatic firmware updates

Complete

- Part of Qcells' complete Q.HOME SMART solution offering
- One brand. One warrantor

qcells

Description

The Q.HOME COMBINER 80 G1 consolidates up to 4 PV strings with a maximum of 44 AC Modules connected into a single enclosure. This enables a smooth interconnection across Q.HOME SMART products while ensuring easy and fast installations. Save time and costs by incorporating this user-friendly combiner box, optimally designed for residential applications. Contact Qcells Customer Support if it is desired to install more than 44 AC Modules.

Technical Specification

GENERAL PRODUCT INFORMATION		Q.HOME COMBINER 80 G1
Manufacturer		Hanwha Solutions Corporation
Product Warranty		5 years
Country of Manufacturer		Vietnam

ACCESSORIES AND REPLACEMENT PARTS		
Supported AC Modules (Microinverter included)		Q.TRON BLK M-G2+/AC
Cellular Modem (LTE-MT-MODEM-CAT4-TN5)		4G based LTE-CAT4 (+5year data plan included)
WiFi Dongle (WIFI-HQ-DG-USB)		FCC Part 15 Subpart C/2412.0 to 2462.0 MHz **
Circuit Breakers		Supports Eaton BR210, BR215*, BR220, BR230, BR240, BR250, and BR260 circuit breakers
Consumption Monitoring CT (CT-JS-CLAMP-200A-5.2m)		A pair of 200 A clamp type current transformers (accuracy ±0.5 %) **

* pre-assembled / ** included in the package
(Others are not included, need to be ordered seperately)

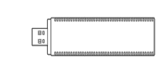
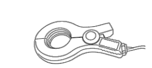
ELECTRICAL SPECIFICATIONS		
System Voltage	[V]	120/240 VAC, 60 Hz
Eaton BR Series Busbar Rating	[A]	125
Max. Continuous Current Rating (input from PV/storage)	[A]	64
Max. Fuse/Circuit Rating (output)	[A]	90
Branch Circuits (solar and/or storage)	[pcs]	Up to four 2-pole Eaton BR series Distributed Generation (DG) breakers only (not included)
Max. Total Branch Circuit Breaker Rating (input)	[A]	80 A of distributed generation/95 A with Gateway breaker included
Gateway Circuit Breaker	[A]	15 A rating Eaton included
Consumption Monitoring	[A]	Metering with a pair of 200 A split core current transformers (accuracy ±2.0 %)
Production Metering	[A]	Metering with 200 A solid core current transformer pre-wired to Gateway (accuracy ±0.5 %)

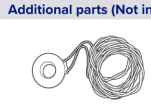
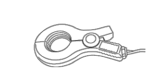
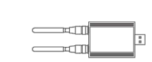
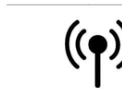
MECHANICAL DATA		
Max. AC Module Connection Q'ty	[pcs]	Up to 44 AC Modules (11 in series × 4 strings)
Dimensions (W × H × D)	[inch]	14.6 × 19.3 × 6.3/height is 21.7 with mounting brackets (37.0 × 49.0 × 16.0 cm/height is 55.1 cm with mounting brackets)
Weights (without connection cables)	[lb]	11.5 (5.2 kg)
Operating Temperature Range	[°F]	−40 to 140 (−40 to 60 °C)
Storage Temperature Range	[°F]	−40 to 140 (−40 to 60 °C)
Enclosure Environmental Rating		Outdoor, NRTL-certified, NEMA type 3R, polycarbonate construction
Wire Sizes		▪ 20 A breaker inputs: 12 to 8 AWG copper conductors ▪ Main lug combined output: 10 to 2/0 AWG copper conductors ▪ Neutral and ground: 8 to 6 copper conductors ▪ Always follow local code requirements for conductor sizing
Cooling		Natural convection
Altitude	[m]	Up to 2,000 (6,561 feet)

INTERNET CONNECTION OPTIONS		
Wi-Fi		IEEE 802.11b/g/n
Cellular		CELLMODEM-CAT4 (4G based LTE-CAT4)
Ethernet		Optional, IEEE 802.3, CAT5E (or CAT6) UTP Ethernet cable

COMPLIANCE		
AC Combiner		▪ UL 1741, CSA C22.2 No.107 ▪ FCC Part 15.B ▪ ANSI C 12.20 accuracy class 0.5 (production meter) ▪ NEMA type 3R ▪ IEEE 2030.5 / CSIP Compliant
Monitoring board		UL 61010-1 / UL 61010-2-030 CSA 22.2 No. 61010-1-12 / CSA 22.2 No. 61010-2-030
CT sensor		Solid core, Split core XOBA

Accessories

Parts included in the package		
 FCC ID: OZ5C307-HW-WF	WIFI-HQ-DG-USB	Wi-Fi dongle with 2.4 GHz bandwidth
 UL2808(XOBA) E498920	CT-JS-CLAMP-200A-5.2 m	A slim clamp type consumption CT with ±0.5 % accuracy (5.2 m, 20 AWG)

Additional parts (Not included in the package)		
 UL2808(XOBA) E535456	CT-HQ-SOLID-200A-2 m	A revenue grade solid core production CT with ±0.1 % accuracy (2 m, 18 AWG) Calibration is required to support revenue grade metering.
 UL2808(XOBA) E498920	CT-JS-CLAMP-200A-25 m	A slim clamp type consumption CT with ±0.5 % accuracy (25 m, 20 AWG)
 UL 62368-1 E150299	LTE-MT-MODEM-CAT4-TN5	Cellular modem with 5 year data plan of AT&T or T-mobile included
	LTE-TN-DP-5Y	5 year data plan extension (for customers previously using cellular option)

Qcells pursues minimizing paper output in consideration of the global environment.

Note: Installation instructions must be followed. Contact our technical service for further information on approved installation of this product.

Hanwha Q CELLS America Inc. 300 Spectrum Center Drive, Suite 500, Irvine CA, 92618 USA | TEL 1-(888) 249-7750 | EMAIL na.support@qcells.com | WEB www.qcells.com/us

qcells

EcoFasten
For Installers. By Installers.

CLICKFIT®

COMPLETE RAIL-BASED RACKING SYSTEM

ClickFit is one of the fastest installing rail-based systems in the industry. Thanks to its Click-In rail assembly, the rails can be connected to any of EcoFasten's composition shingle, tile, and metal roof mounts in seconds without the need for fasteners or tools. The ClickFit system is made of robust materials such as aluminum and coated steel, to ensure corrosion-resistance and longevity. ClickFit conforms to UL 2703 and has been tested in extreme weather conditions including wind, fire, and snow.

FEATURES & BENEFITS

- Pre-installed rail fastening bolt
- Fully integrated bonding
- Click-On Mid & End Clamps
- Compatible with a variety of EcoFasten roof attachments

ROOF TYPE: Composition Shingle, Tile, Metal

RACKING: Rail-Based

ATTACHMENT: Structural-Attach Direct-Attach

FAST INSTALLING SYSTEM FEATURING CLICK-IN RAIL ASSEMBLY

INTERNAL SPLICE

Tool-free bonded internal Splice installs in seconds.

CF MLPE MOUNT

Attach Module Level Power Electronics to the top of the rail.

MID CLAMP

Click-on mid clamp features integrated bonding pins and fits module frames sized 30-50mm.

END CLAMP

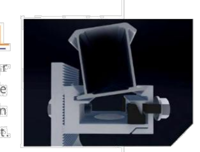
Click-on end clamp fits module frames sized 30-50mm.

END CAP

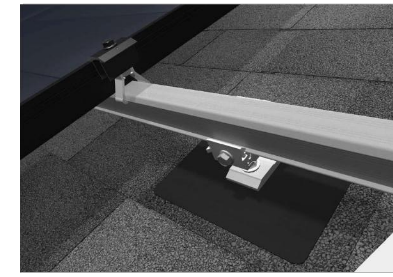
The slide-on end caps allow the end clamps to be accurately positioned on the rail in seconds, while providing an aesthetically pleasing, finished install.

RAIL

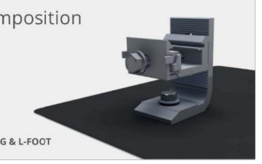
The ClickFit rail clicks into our proprietary composition shingle & tile L-foot and is tightened in place with a pre-installed bolt.



COMPOSITION SHINGLE



Combine the versatile ClickFit L-Foot with the watertight GF-1 flashing for a fast installation on composition shingle roofs.



GF-1 FLASHING & L-FOOT

TILE ROOFS

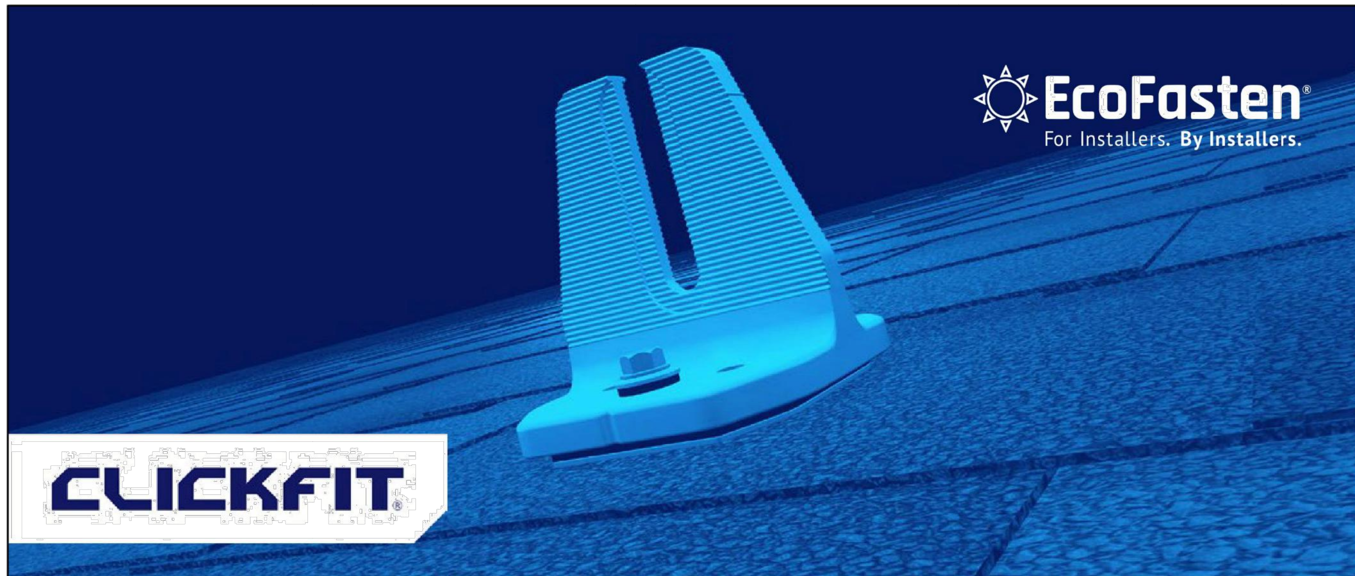
CLICKFIT TILE HOOK

Use the adjustable ClickFit Tile Hook for attaching the ClickFit system to tile roofs. Works with Flat, S, and W tile profiles.

STANDING SEAM METAL ROOFS

Combine the ClickFit L-Foot with SimpleBlock®-U for a fast installation on standing seam metal roofs.

SIMPLEBLOCK-U



CLICKFIT SMART FOOT

Introducing our fast-installing Smart Foot! Created for use with our rail-based ClickFit System, the mount easily attaches onto composition shingle roofs. Fabricated from cast aluminum, the ClickFit Smart Foot features our proprietary UltraGrip Technology™. A layer of foam cushioning helps embed the waterproofing sealant deep into the shingle's granules and flexibly conforms over the steps found on architectural-style shingles. Simply peel the backing off the release liner from the underside of the UltraGrip pad and place the mount on the shingles. The mount comes packaged with the ClickFit "Clicker". Screws are sold separately. ClickFit System conforms to UL 2703, UL 2703A and UL 3741.

Features & Benefits

- Integrated flashing utilizes **UltraGrip Technology™** to create a watertight seal
- Multiple opportunities to find the rafter
- Eliminates the need to drill pilot holes
- Most installations require no sealant
- No need to add adhesive shims for shingle leveling
- Built-in alignment guide for fast "Clicker" leveling

CLICKFIT SMART FOOT

Integrated UltraGrip Technology™

ClickFit Smart Foot comes with pre-installed sealing pads that are compatible with all composition shingle roofs. These pads can be installed in temperatures as low as 5° F, and when fastened to the roof, create a strong and watertight seal. In most cases, there's no need for additional sealant when mounting Smart Foot to the roof deck. The layer of flexible foam on the sealing pads provides cushioning, allowing the super-sticky waterproofing sealant to embed deep into the shingle's granules, conforming to the steps on architectural-style shingles. The foam also works with the bonded sealing washer to absorb daily movements caused by thermal expansion and contraction, preventing any damage to the shingles.


A layer of flexible foam pushes the sealant deep into the shingle's granules and around shingle steps


ClickFit Smart Foot can be installed on top of architectural shingles steps

Testing & Documentation

- Florida Product Approval
- ClickFit Installation Manual
- ClickFit CutSheets
- ClickFit Parts List

ULTRAGRIP TECHNOLOGY

WATER TIGHT TECHNOLOGY

EcoFasten
AN ENSTALL COMPANY
4141 W. VAN BUREN ST, SUITE 2, PHOENIX AZ 85009
1-877-859-3947 | INFO@ECOFASTENSOLAR.COM

GODWIN ENGINEERING AND DESIGN, LLC

8378 Foxtail Loop, Pensacola, FL 32526 | (850)712-4219 | chad@godwineng.com

March 25, 2026

To: Hernando County Building Division
789 Providence Blvd
Brooksville, FL 34601

Subject: Aragones - Residential PV Roof Mount Installation
151 Borland St
Brooksville, FL 34604

To whom it may concern,

This letter is regarding the proposed installation of a rooftop-mounted Solar PV system on the existing residential structure at the subject address. I have reviewed the attachment plan and have determined that the rooftop-mounted PV system is in compliance with the applicable sections of the following Codes as amended and adopted by the jurisdiction when installed in accordance with the manufacturer's installation instructions:

2023 Florida Building Code 8^{th} Edition, FBC
ASCE 7 Min. Design Loads for Buildings & Other Structures
Design Criteria: Design Wind Speed(Vult) - 140 mph 3sec gust, Exposure Category – C, Risk Category II

The rooftop-mounted photovoltaic panel system has been designed in accordance with FRC R324.4. When roof penetrations are necessary, they shall be flashed and sealed in accordance with the manufacture's installation instructions, R905.17.3. The PV system consist of the modules, railing, and connection hardware. Refer to the specific roof type calculation pages for PV dead loads. For the portions of the roof with PV installed, the existing roofing system provides an adequate load path for transferring the concentrated reactions to the building frame and the structure will accommodate the additional gravity and environmental loads from the PV system installation per FBC 1607.13.4.

The securement method of the the PV system is to be mounted parallel to the structure with the site specific railing and attachments according to the designed plans. The site specific wind load calculations for the module and their supports are attached with this document. Fasteners shall be installed to the designated roof member with the proper torque from the manufactures installation instructions.

The design wind pressures for rooftop solar panels located on enclosed or partially enclosed buildings of all heights, with panels parallel to the roof surface with a tolerance of 2° and with a max height above the roof surface, h_2, not exceeding 10 in. A min gap of 0.25 in shall be provided between all panels with the spacing of gaps between panels not exceeding 6.7 ft. in addition the array shall be located at least $2h_2$ from the roof edge, a gable ridge, or a hip ridge.

It is the contractors responsibility to review all drawings for accuracy and notify the EOR of any discrepancies prior to beginning construction. To the best of my knowledge, the plans and specifications comply with the minimum requirements of the latest Florida Building code.

Please see attached documents and contact me should you have any questions.

Sincerely,
D. Chad Godwin, PE 81360
Exp. 02/28/2027

D. CHAD GODWIN
LICENSE
No 81360
STATE OF
FLORIDA
PROFESSIONAL ENGINEER

GODWIN ENGINEERING AND DESIGN, LLC

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Roof Structure Details		Aragones - Residential Calculations Sheet - R1-R2
Roof Angle	8° to 20°	The securement method of the PV system is to be mounted parallel to the Asphalt Shingle roof with the Clickfit railing and Eco Clickfit Smartfoot flashings/attachments. The mounts should be staggered, where possible, to allow distribution of the design loads evenly to the structure. The mounts shall be installed with (2) #14 Self-Dr x 3" W/BW, to Truss
Roof Type	Hip	
Roof Covering	Asphalt Shingle	
Mean Roof Height	15 ft	
Truss Spacing	24 in O.C.	
Rafter/Truss Size	2 x 4	

Wind Load Paramaters

Wind Speed (asd)		108	mph	FRC R301.2.1.3	Basic Wind Speed (Ult)	140	mph
Effective Wind Area		21.02	ft^2	26.2	Exposure Cat.	C	B,C, or D
Wind Directionality	K_d	0.85		Table 26.6-1	Elevation	<1000	ft
Topographic factor	K_{zt}	1.00		26.8 or 26.8.2	bldg. least hori. dim (typ.)	360	in
Ground Elevation Factor	K_e	1.00		Table 26.9-1	Roof Height, *h*	15.00	ft
Velocity Exposure Coefficient	K_z	0.85		Table 26.10-1			
Array Edge Factor	γ_E	1.50	*Exposed*	29.4.4			
Array Edge Factor	γ_E	1.00	*Non. Exp*	29.4.4			
Solar Panel Equalization Factor	γ_a	0.67		Fig. 29.4-8(gap = 0.25")			
Velocity Pressure	q_h	25.59	psf	q_h=0.00256 $K_zK_{zt}K_eV^2$			
Added Safety Factor		1					
Allowable Pullout per mount		1004.0	lbs				
	0.4h or 0.6h	6.00	ft	Flat - 0.6h, Gab/Hip - 0.4h			
10% of least horizontal dim		3.00	ft	10% of least hor. Dim. Or 0.4h, whichever is smaller, but not less than either 4% of Least hor. Or 3ft. (flat roof - 0.6h)			
Roof Zone Set Back	a	4.00	ft				
	h 2	6	in	Not > 10in(panel height above roof)			
	2*h* 2	12	in	*min distance array shall be from the roof edge, Gable Ridge, or hip ridge			
		0.25	in	min gap between all panels but not > 6.7ft			
	d1	1.00	ft	Horizontal distance orthogonal to panel edge			
	d2	0.25	ft	Horizontal distance from edge of one panel to the nearest edge in the next row			
	0.5*h*	7.50	ft				

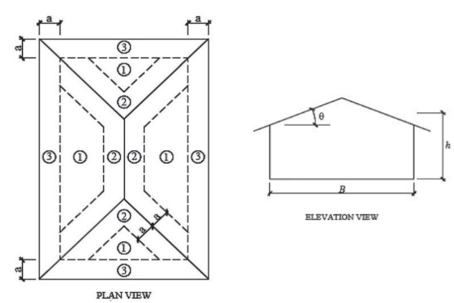
Exposed Module Definition

Exposed factor = 1.5 for uplift loads on panels that are exposed and within a distance of 2*h* 2 from an edge of the array. Modules are condidered Exposed if d_1 to the roof edge > 0.5*h* and one of the following applies:

1. d_1 to adjacent array > 2*h* 2.
2. d_2 to the next adjacent panel > 2*h* 2.

PV Attachment - Results

Exposed	R1-R2 Roof Zones - Hip 8° to 20°		
Non-Exp.	1	2	3
GC_p - Uplift	-1.56	-2.13	-2.3
GC_p - Down	0.57	0.57	0.57
$p=q_hK_d(GC_p)(\gamma_E)(\gamma_a)$ (psf) ASCE 29.4-7	-31.1	-43.6	-47.6
	-19.8	-28.1	-30.7
	16.0	16.0	16.0
	16.0	16.0	16.0
PL(lb) (Portrait Rails) = p * A_{eff}	-352	-492	-269
	-223.5	-317.1	-347.5
PL(lb)(landsc. Rails) = p * A_{eff}	-231.8	-324.3	-354.2
	-147.2	-208.8	-228.8
Mx. Span (in) (Portrait)	48	48	24
	48	48	48
Mx. Span(in) (Landsc.)	48	~~48~~	~~24~~
	48	48	48
Cantilever (Portrait) Span * 33%	15.8	15.8	7.9
	15.8	15.8	15.8
Cantilever (Landsc.) Span * 33%	15.8	15.8	7.9
	15.8	15.8	15.8



a = 4ft ; 10% of least hor. Dim. Or 0.4h, whichever is smaller, but not less than either 4% of Least hor. Or 3ft. (flat roof - 0.6h)

B - Horizontal dimension of bldg. normal to wind direction, (ft)

h = Mean roof height, in (ft), except that eave height shall be used for θ ≤ 10°.

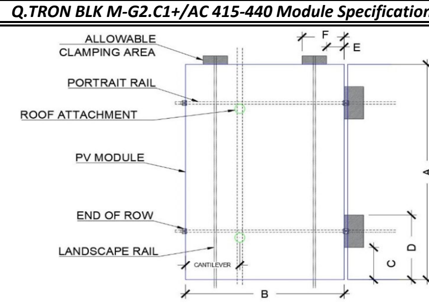
θ = Angle of Plane of roof from horizontal, degrees.

***** Spans with ~~Mark through~~ denote allowable Module pressure rating is exceeded. 3 Rail is required**

PV Dead Load

QTY of Modules (15 in Portrait,)			15
Module Area		21.02	ft^2
Rail, Clamps, Mounts		1	lb/ft
Total Rail Length		114	ft
Module	W_{mod}	51	lbs
Array	W_{mods}	759	lbs
Micro/optimizer	W_{mic}	60	lbs
PV Rail	$W_{PV\ rail}$	114	lbs
Total Weight	W_{total}	932	lbs
Total Area	A_T	315.29	ft^2
Dead Load	D_{PV}	2.96	psf
Weight/attachment		24.5	lbs

Q.TRON BLK M-G2.C1+/AC 415-440 Module Specifications



A (ft)	B (ft)	C (in)	D (in)	E (in)	F (in)
5.65	3.72	0.79	11.81	0.79	7.87

Fastener Allowable Pullout

(2) #14 Self-Dr x 3" W/BW

Diameter	#N/A	
S.G.	#N/A	
Thread Embedment per	#N/A	in
# of Fastener	#N/A	
Pullout Value (Source - FL Product Approval)	1004.0	lb_f

Module load ratings (psf)	Ultimate	Allowable(Ult /1.5)	
Load Rating - Snow	113.4	75.6	Portrait
Load Rating - Wind	-113.4	-75.6	
Load Rating - Snow	50.4	33.6	Landscape
Load Rating - Wind	-50.4	-33.6